

Bineta DABO

Researcher in Distributed Systems & Cloud Computing — PhD application

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Master's graduate in Distributed Information Systems (UCAD, with honours), author of a thesis on multi-agent Cloud resource allocation, published on HAL and awarded 1st prize at the international conference COC'2026. Hands-on experience with multi-tenant distributed systems under strong consistency and scalability constraints. Interests: fairness and efficiency of distributed resource allocation, multi-agent systems, Cloud Computing.

RESEARCH & ACADEMIC PROJECTS

- Multi-agent model for fair Cloud resource allocation** · Master's thesis, UCAD Defended Jul. 2025 · 16/20
- Multi-agent system (Python, SPADE, asyncio): client, resource, load-balancing and monitoring agents; monitoring dashboard.
 - HRRN priority with aging (anti-starvation) and topological-sort scheduling (Kahn's algorithm): fairness and respect of task dependencies.
 - Published on HAL (hal.science/hal-05571848) · 1st prize at the international conference COC'2026.

Cloud Computing — AWS, OpenStack & CloudSim · Master 2 SIR, UCAD

- Bash / Java (AWS SDK) automation of an AWS VPC; CloudSim simulations (200 to 1000 requests): impact of the number of VMs on response time.

PROFESSIONAL EXPERIENCE

- Full-Stack Developer** · IAI Compétences, Dakar Apr. 2026 – present
- Logistics platform (Node.js / Express / PostgreSQL) for a port operator in Dakar; fixed critical bugs (cascading 503s, stock recalculation, orders API).
 - Multi-clinic medical platform (React/Vite, Supabase): multi-tenant isolation with Row Level Security; ~100 SQL tables, ~83 commits.

- Backend Developer — Mawlid, Pilgrim Guide (Gamou de Tivaouane, CEZAT)** 2026
- Backend developer of the team (official app): PostgreSQL, RPC, Edge Functions, role-based security (RLS), realtime; k6 Cloud load testing and production incidents.

- Freelance Web Developer** · Etech Energie Feb. 2026 – present
- Design and development of the company website.

- QA Tester (3-month internship)** · SecurAxe SUARL, Dakar Jun – Aug 2026
- Functional testing of the Archiva web platform and the Kaaysign mobile app.

EDUCATION

- Master's (M2) Computer Science — Distributed Information Systems** · UCAD, Dakar 2023–2025 · defended Jul. 2025 · With honours

- Master's (M1) Computer Science** · UCAD, Dakar 2022–2023

- Bachelor's (L3) Computer Science** · UCAD, Dakar 2021–2022

- BTS Industrial Computing & Networks** · CFPT Senegal-Japan, Dakar 2020–2021

TECHNICAL SKILLS

- Backend:** Node.js / Express, Supabase (Auth, RPC, Edge Functions, Realtime, Storage), Java / Spring Boot, ASP.NET Core, Laravel, PHP
- Frontend & mobile:** React 18/19, Vite, Next.js, Tailwind CSS, Three.js / R3F, Recharts, FullCalendar, Bootstrap, Flutter, Expo / React Native
- Databases:** PostgreSQL, MySQL, MongoDB, Couchbase, Oracle, SQL Server
- Cloud & infra:** AWS (EC2, S3, VPC), OpenStack, CloudSim, Docker, Vercel, Linux, Bash, networking (TCP/IP, DNS, DHCP, routing)
- Data & AI:** Python, NumPy, Pandas, R, Machine Learning, Deep Learning, multi-agent systems (SPADE)
- Tools & methods:** Git / GitHub, CI/CD (Jenkins, GitLab CI), k6, Jira, Trello, ClickUp, UML, BPMN, Scrum

CERTIFICATIONS

- AfriConnect+ / Erasmus+ (2025–2026): IoT security (TLS/DTLS, VPN), embedded systems and industrial IoT, LoRaWAN, FreeRTOS on STM32L4, data mining and AI, IoT data analysis with R, PCB design in KiCad.

LANGUAGES

- French: fluent · Wolof: fluent · English: advanced reading, intermediate writing, speaking improving